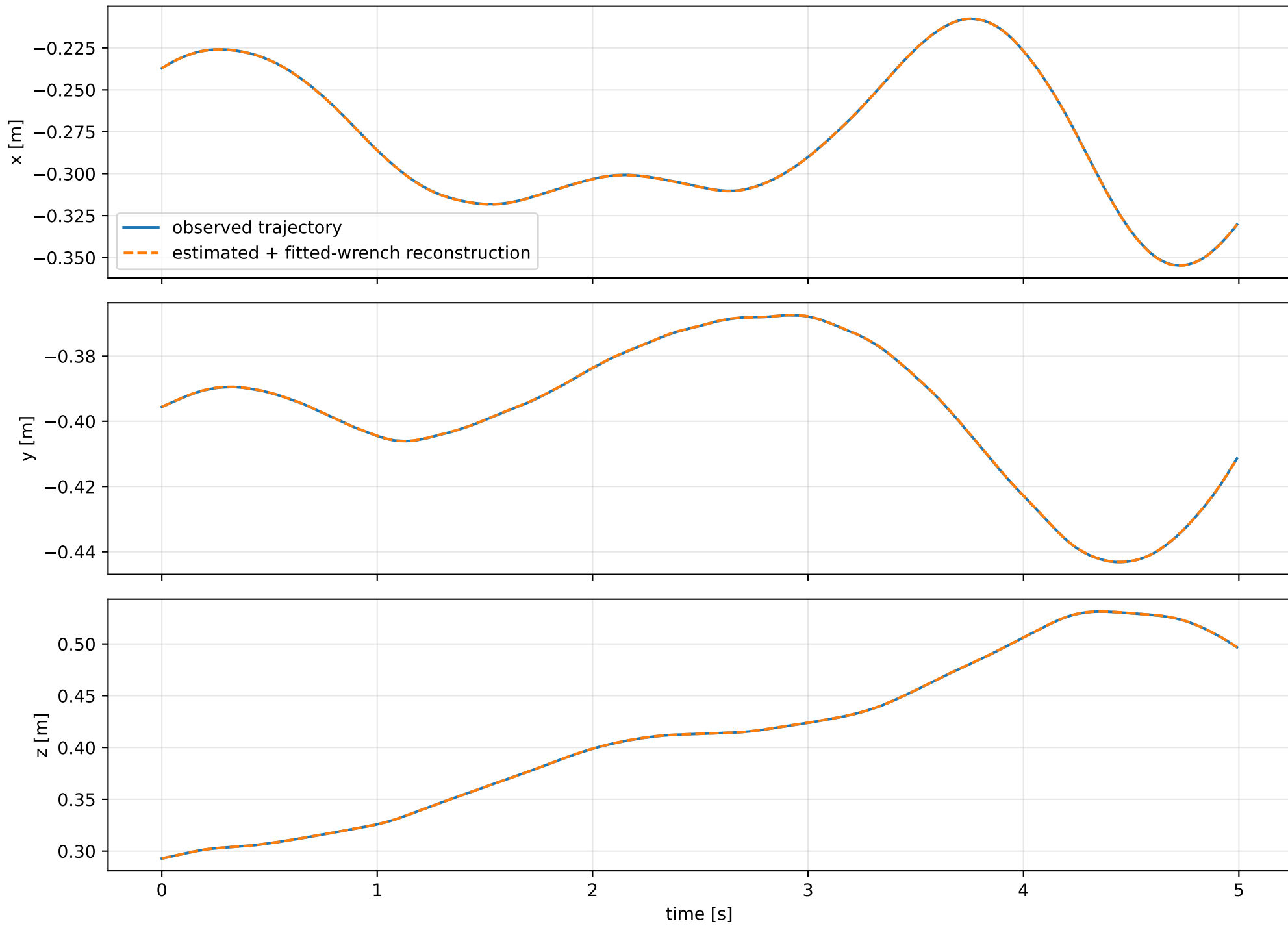
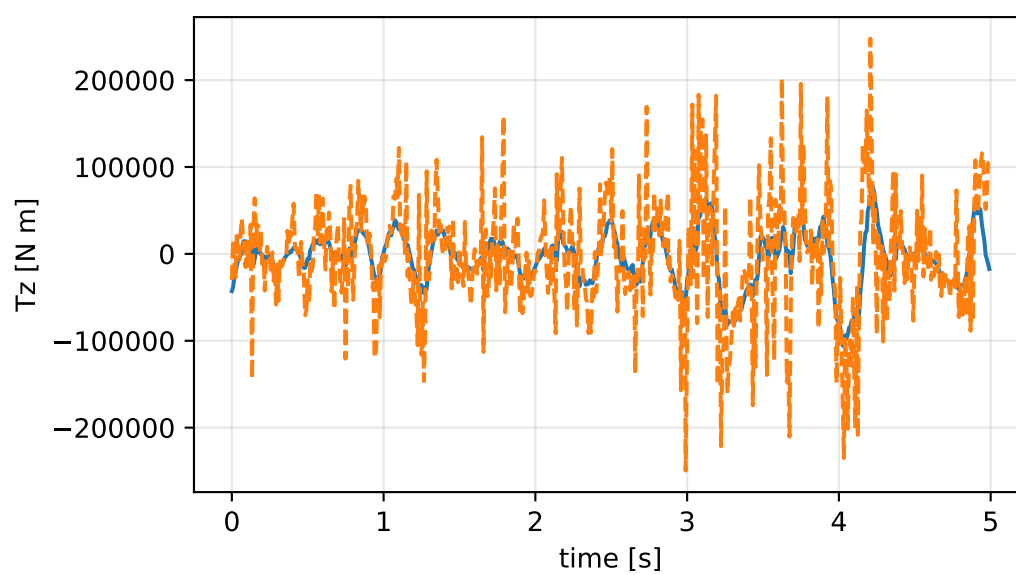
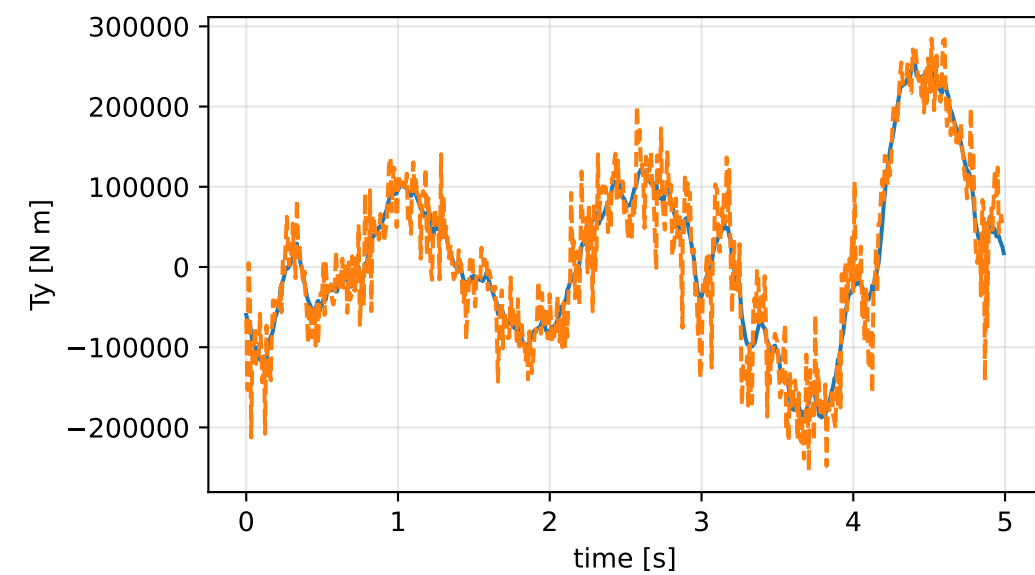
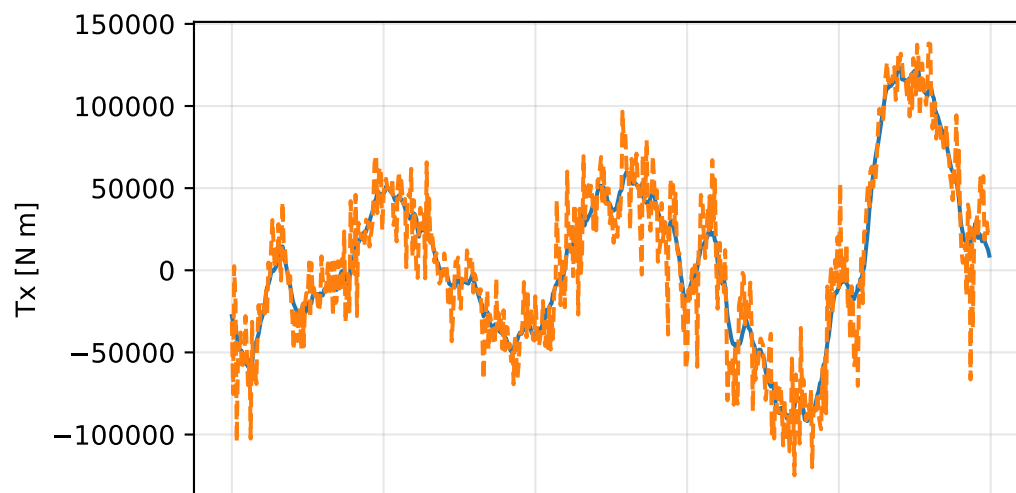
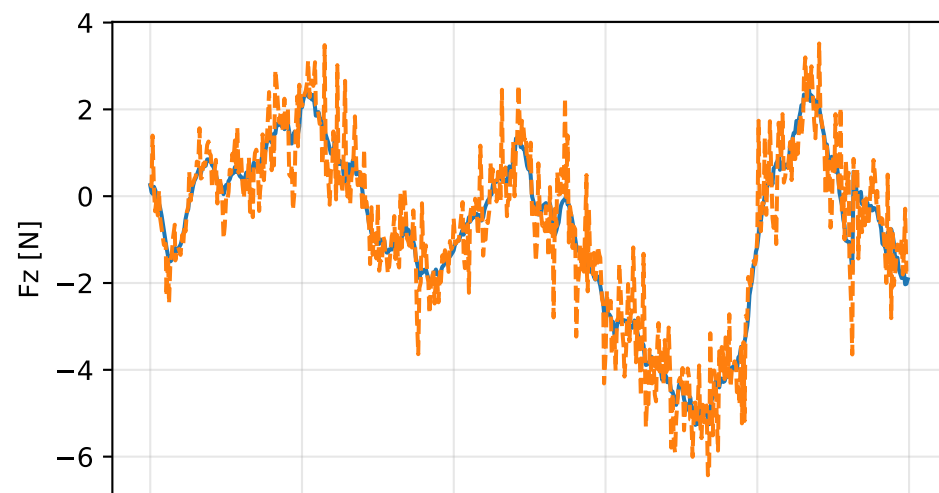
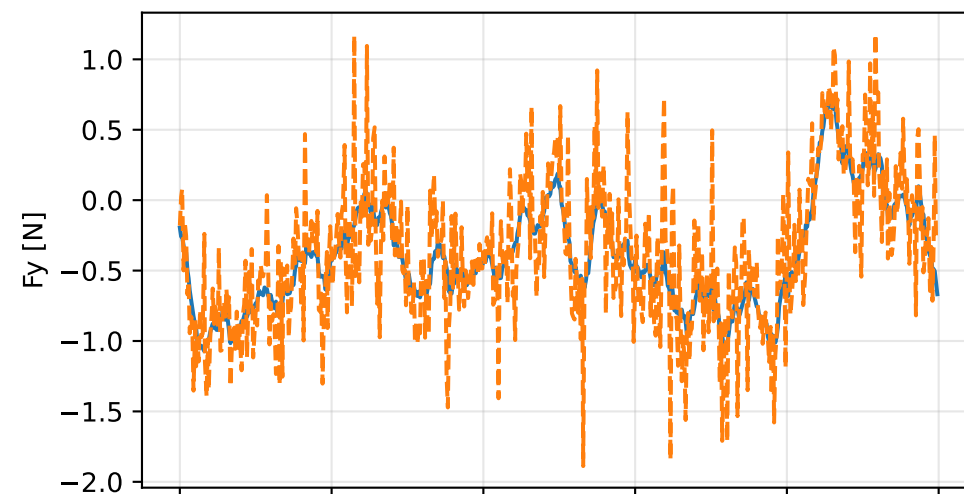
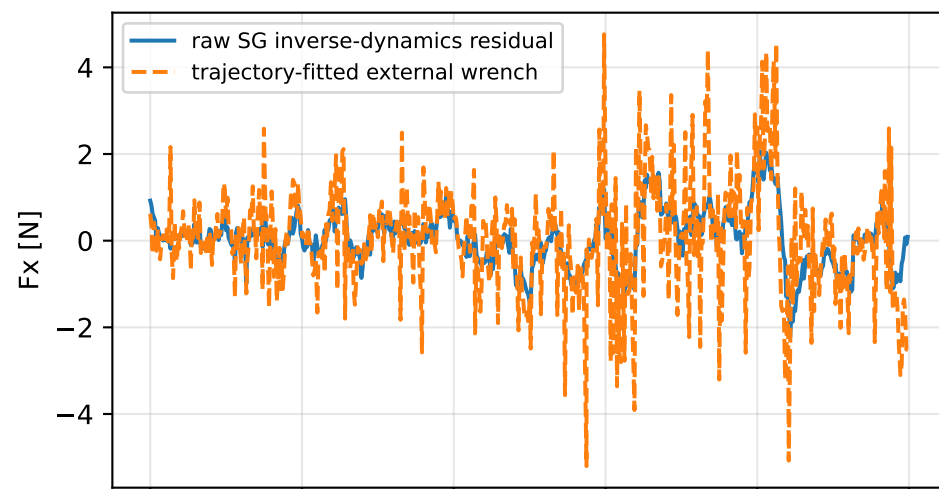


jacobian_finite_difference: trajectory

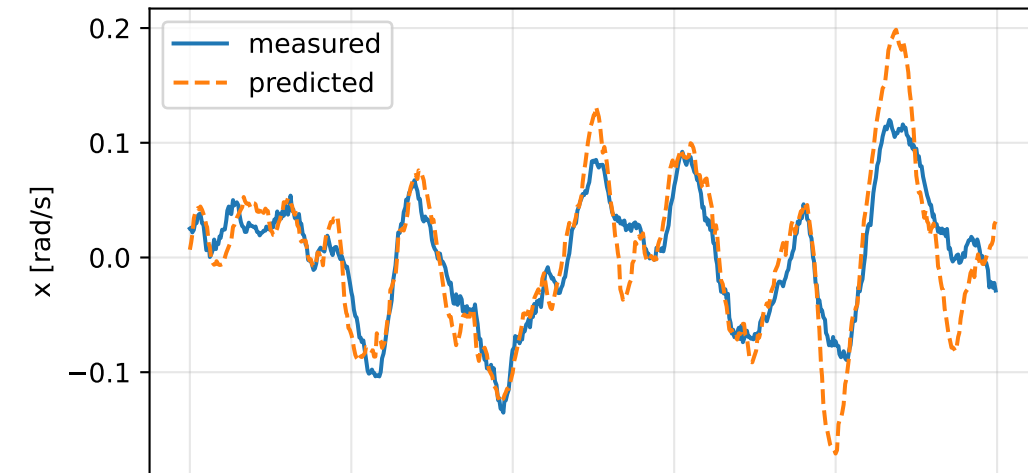


jacobian_finite_difference: residual wrench

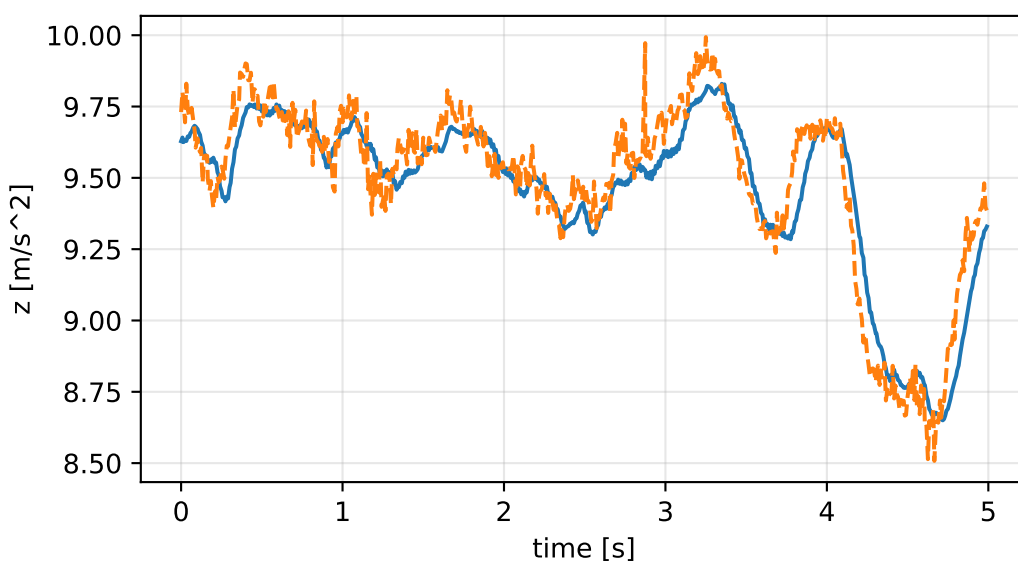
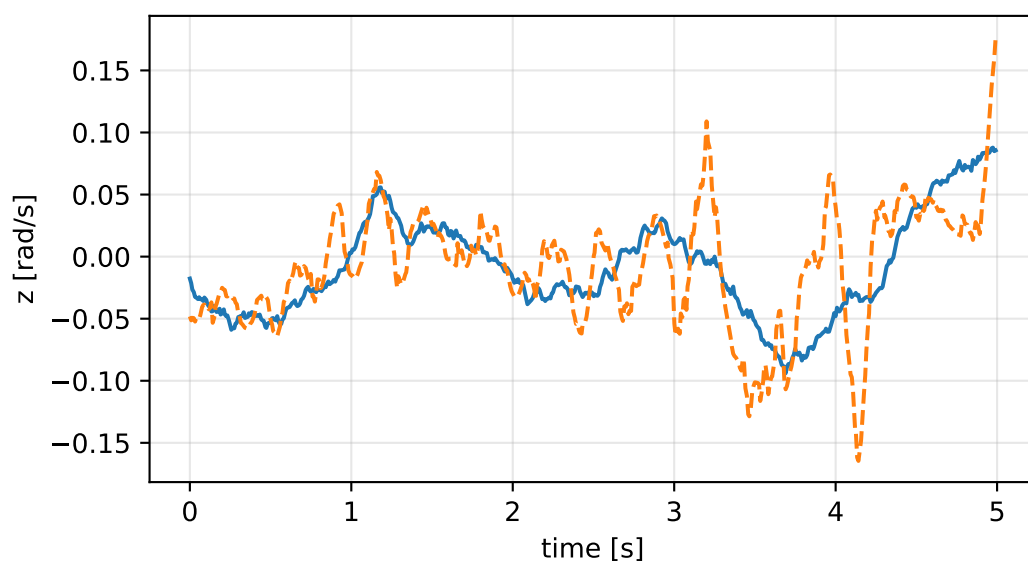
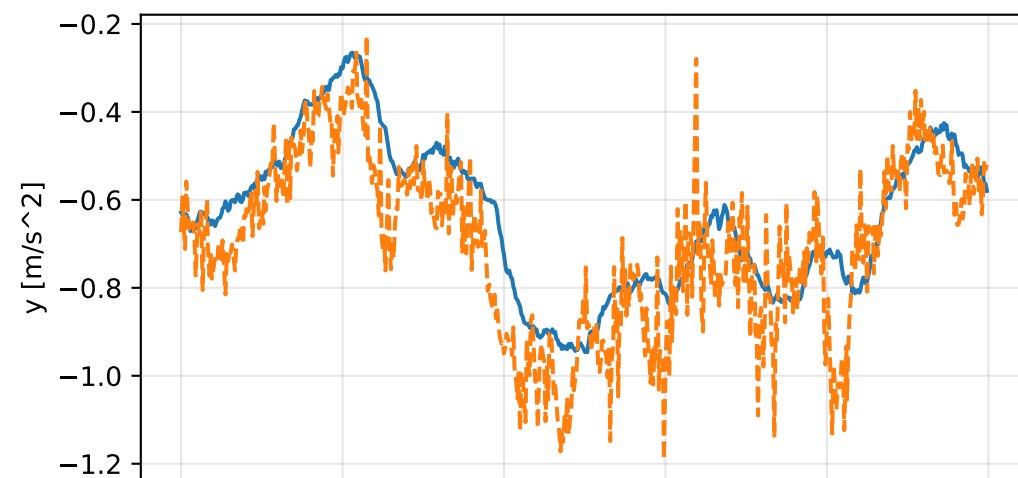
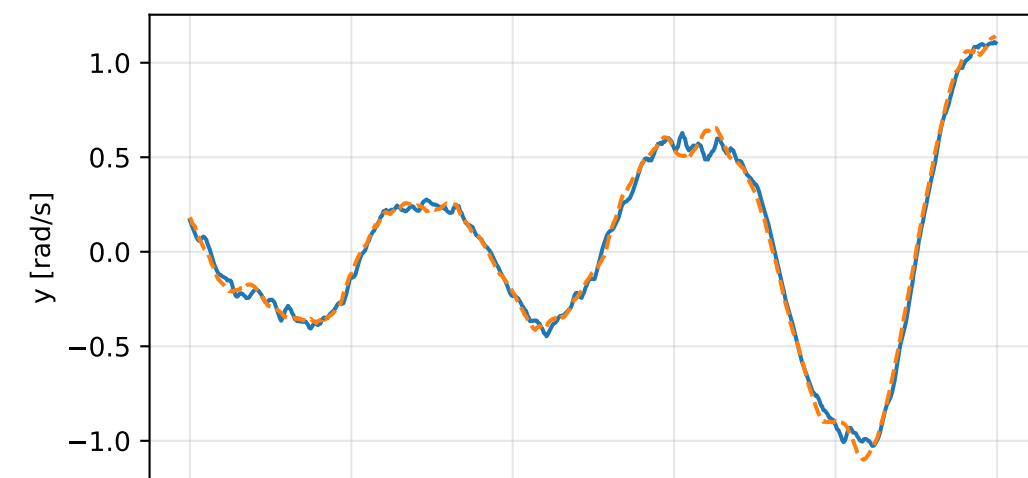
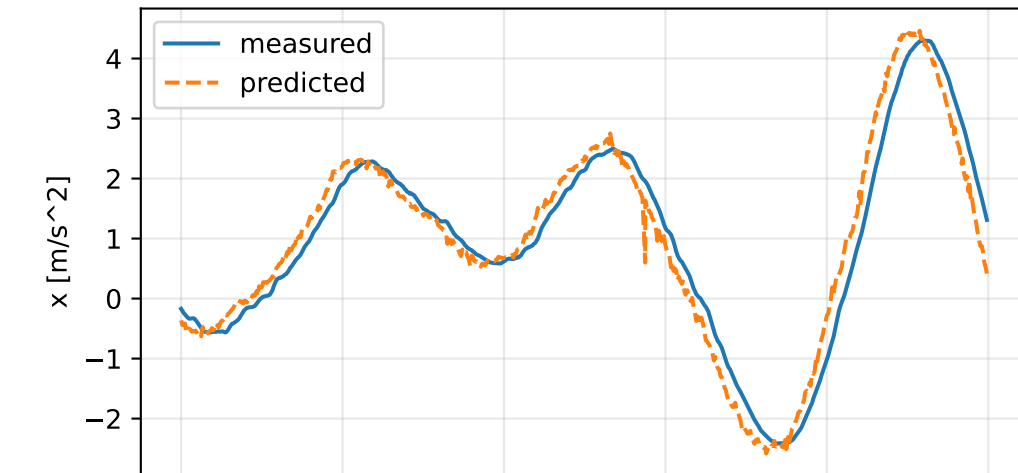


jacobian_finite_difference: measured sensor comparison

gyro



specific force



jacobian_finite_difference: nominal -> estimated parameters

Case: jacobian_finite_difference

status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 4.66206763 delta 2.31047004

inertia [kg m²] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [13626.2023271 27822.5906286 -1723.6236809] d=[13626.137327 27822.5906286 -1723.6236809]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [27822.5906286 57079.7857825 840.8927836] d=[27822.5906293 57079.7208125 840.8927836]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-1723.6236809 840.8927836 70601.7558365] d=[-1723.6236999 840.8927836 70601.7558365]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [-0.0890193 0.2009899 -0.0259735] d=[-0.0869946 0.2010204 -0.0259735]

rotor force effectiveness

[1. 1. 1. 1.] -> [7.9760006e-06 2.0327264e+00 3.4911494e+00 1.4485878e+00] d=[-0.999992 1.0327264 2.4911494 0.4485878]

rotor lag [s] 0.103122841

gimbal lag [s] 0.0403628947